Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech | | ***Time: 03 Hours***
Code: 100311 | ***Subject: Mathematics-III (Differential Calculus)*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) The solution of the differential equation $y'' - 3y' + 2y = e^{3x}$

(i) $C_1e^{x} + C_2e^{2x} + \frac{e^{3x}}{2}$ (ii) $C_1e^{-x} + C_2e^{-2x} + \frac{e^{3x}}{2}$
(iii) $C_1e^{-x} + C_2e^{2x} + \frac{e^{3x}}{2}$ (iv) None of these

(b) The solution of $p + q = z$ is

(i) $f(xy, y\log z) = 0$ (ii) $f(x + y, y + \log z) = 0$
(iii) $f(x - y, y - \log z) = 0$ (iv) None of these

(c) The series $1 - \frac{1}{(1!)^2}\left(\frac{x}{2}\right)^2 + \frac{1}{(2!)^2}\left(\frac{x}{2}\right)^4 - \frac{1}{(3!)^2}\left(\frac{x}{2}\right)^6 - \cdots$ is

(i) $J_{1/2}(x)$ (ii) $xJ_{1/2}(x)$
(iii) $xJ_0(x)$ (iv) None of these

(d) If $\int_{-1}^{1} P_n(x)\,dx = 2$, then n is

(i) 0 (ii) 1
(iii) 2 (iv) None of these

(e) The particular solution of $y'' + 9y = \sin 3x$ is

(i) $-\frac{x}{6}\cos 3x$ (ii) $\frac{x}{6}\cos 3x$
(iii) $-\frac{3x}{2}\cos 3x$ (iv) $\frac{3x}{2}\cos 3x$

(f) The differential equation $y'' + (s^3 \sin x)^5 y' + y = \cos x^3$ is

(i) Homogeneous (ii) Non-linear
(iii) Second order linear (iv) Non-homogeneous with constant coefficient

(g) The magnitude of the directional derivative of the function $f(x, y) = x^2 + 3y^2$ in a direction normal to the circle $x^2 + y^2 = 2$, at the point (1,1) is

(i) $4\sqrt{2}$ (ii) $5\sqrt{2}$
(iii) $7\sqrt{2}$ (iv) $9\sqrt{2}$

(h) If $\vec{R} = xi + yj + zk$ is the position vector, then $div\,\vec{R}$ is

(i) 1 (ii) 3
(iii) -3 (iv) -2

(i) The degree of the homogeneous function $f(x, y) = \frac{x^{[illegible]} + y^{[illegible]}}{x+y}$ is

(i) 4/5 (ii) 4/3
(iii) 7/3 (iv) None of these

(j) If $f(x,y) = \tan^{-1} y/x + \sin^{-1} x/y$, then $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} =$

(i) 2 (ii) 1
(iii) 0 (iv) -1

Q.2 (a) If $y^{1/m} + y^{-1/m} = 2x$, prove that [7]
(i) $(x^2 - 1)y_2 + xy_1 - m^2 y = 0$
(ii) $(x^2 - 1)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0$

(b) If $p^2 = a^2\cos^2\theta + b^2\sin^2\theta$, prove that $p + \frac{d^2p}{d\theta^2} = \frac{a^2b^2}{p^3}$ [7]

Q.3 (a) Use Taylor's theorem to expand $f(x,y) = x^y$ in powers of $(x-1)$ and $(y-1)$. [7]
(b) Find the dimension of the rectangular box, open at the top, of maximum capacity whose surface area is 432 sq.cm. [7]

Q.4 (a) If $u = \tan^{-1}\left(\frac{y^2}{x}\right)$, prove that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -\sin^2 u . \sin 2u$ [7]

(b) If by the substitution $u = x^2 - y^2, v = 2xy, f(x,y) = \theta(u,v)$, show that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 4(x^2 + y^2)\left(\frac{\partial^2\theta}{\partial u^2} + \frac{\partial^2\theta}{\partial v^2}\right)$ [7]

Q.5 (a) Verify Stoke's theorem for $F = (2x - y)I - yz^2J - y^2zK$ over the upper half surface $x^2 + y^2 + z^2 = 1$, bounded by its projection on the xy −plane. [7]
(b) If the directional derivative of $\varphi = ax^2y + by^2z + cxz^2$ at the point $(1,1,1)$ has maximum magnitude 15 in the direction parallel to the line $\frac{x-1}{2} = \frac{y-3}{-2} = z$, find the values of a, b, c [7]

Q.6 (a) Solve $(x^2 - y^2)(xdx + ydy) = a^2(xdy - ydx)$ [7]
(b) Solve $p(p + x) = y(x + y)$ [7]

Q.7 (a) Solve $(2y\log y + y - x)dy = ydx$ [7]
(b) Solve, by the method of variation of parameters, $y'' - 2y' + y = e^x \log x$ [7]

Q.8 (a) Solve $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + y = \log x\frac{\sin(\log x)+1}{x}$ [7]
(b) Prove that $J_m(x) = \frac{1}{\pi}\int_0^\pi \cos(m\theta - x\sin\theta)\,d\theta$, where '$n$' is positive integer [7]

Q.9 (a) Solve the equation : $(z^2 - 2yz - y^2)p + (xy + zx)q = xy - zx$ [7]
(b) Solve the equation : $(x - y)(px - qy) = (p - q)^2$ [7]